

# Lab03\_Franco

AUTHOR

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```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.setenv(CHROMOTE_CHROME_ARGS = "--no-sandbox --disable-gpu")

cat("Arquivo compilado em:", format(Sys.time(), "%d/%m/%Y às %H:%M:%S (%Z)"))
```

Arquivo compilado em: 01/09/2026 às 09:10:44 (-03)

## Part 1

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

— Column specification —

Delimiter: ","

chr (2): IATA\_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airports <- read_csv(
  "airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)
)
```

Rows: 322 Columns: 5

— Column specification —

Delimiter: ","

chr (3): IATA\_CODE, CITY, STATE

dbl (2): LATITUDE, LONGITUDE

- Use ``spec()`` to retrieve the full column specification for this data.
- Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
processa_bloco <- function(bloco, pos) {
  bloco %>%

  filter(!is.na(DESTINATION_AIRPORT) & !is.na(ARRIVAL_DELAY)) %>%

  filter(!str_starts(DESTINATION_AIRPORT, "1")) %>%

  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    soma_atraso = sum(ARRIVAL_DELAY),
    n_voos = n(),
    .groups = "drop"
  )
}

processa_bloco <- function(bloco, pos) {
  bloco %>%
  filter(!is.na(DESTINATION_AIRPORT) & !is.na(ARRIVAL_DELAY)) %>%
  filter(!str_starts(DESTINATION_AIRPORT, "1")) %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    soma_atraso = sum(ARRIVAL_DELAY),
    n_voos = n(),
    .groups = "drop"
  )
}

estatisticas_bloco <- read_csv_chunked(
  "flights.csv",
  callback = DataFrameCallback$new(processa_bloco),
  chunk_size = 1000000,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

flights <- estatisticas_bloco %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    ARRIVAL_DELAY = sum(soma_atraso) / sum(n_voos),
    .groups = "drop"
  )
```

## Part 2

```
aeropertos_ausentes <- flights %>%
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

aeropertos_ausentes
```

# A tibble: 0 × 2

# i 2 variables: DESTINATION\_AIRPORT <chr>, ARRIVAL\_DELAY <dbl>

```
flights %>%
  left_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

# A tibble: 322 × 6

|    | DESTINATION_AIRPORT | ARRIVAL_DELAY | CITY          | STATE | LATITUDE | LONGITUDE |
|----|---------------------|---------------|---------------|-------|----------|-----------|
|    | <chr>               | <dbl>         | <chr>         | <chr> | <dbl>    | <dbl>     |
| 1  | ABE                 | 5.80          | Allentown     | PA    | 40.7     | -75.4     |
| 2  | ABI                 | 4.08          | Abilene       | TX    | 32.4     | -99.7     |
| 3  | ABQ                 | 5.61          | Albuquerque   | NM    | 35.0     | -107.     |
| 4  | ABR                 | -3.39         | Aberdeen      | SD    | 45.4     | -98.4     |
| 5  | ABY                 | 8.68          | Albany        | GA    | 31.5     | -84.2     |
| 6  | ACK                 | 4.91          | Nantucket     | MA    | 41.3     | -70.1     |
| 7  | ACT                 | 10.6          | Waco          | TX    | 31.6     | -97.2     |
| 8  | ACV                 | 9.68          | Arcata/Eureka | CA    | 41.0     | -124.     |
| 9  | ACY                 | 11.8          | Atlantic City | NJ    | 39.5     | -74.6     |
| 10 | ADK                 | -3.19         | Adak          | AK    | 51.9     | -177.     |

# i 312 more rows

```
flights <- flights %>%
  left_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

```
head(flights)
```

# A tibble: 6 × 6

|   | DESTINATION_AIRPORT | ARRIVAL_DELAY | CITY        | STATE | LATITUDE | LONGITUDE |
|---|---------------------|---------------|-------------|-------|----------|-----------|
|   | <chr>               | <dbl>         | <chr>       | <chr> | <dbl>    | <dbl>     |
| 1 | ABE                 | 5.80          | Allentown   | PA    | 40.7     | -75.4     |
| 2 | ABI                 | 4.08          | Abilene     | TX    | 32.4     | -99.7     |
| 3 | ABQ                 | 5.61          | Albuquerque | NM    | 35.0     | -107.     |
| 4 | ABR                 | -3.39         | Aberdeen    | SD    | 45.4     | -98.4     |
| 5 | ABY                 | 8.68          | Albany      | GA    | 31.5     | -84.2     |
| 6 | ACK                 | 4.91          | Nantucket   | MA    | 41.3     | -70.1     |

## Part 3

```
aeropertos_por_estado <- airports %>%
  count(STATE, name = "num_aeropertos", sort = TRUE)

aeropertos_por_estado
```

```
# A tibble: 54 × 2
  STATE num_aeroportos
  <chr>      <int>
1 TX          24
2 CA          22
3 AK          19
4 FL          17
5 MI          15
6 NY          14
7 CO          10
8 MN           8
9 MT           8
10 NC           8
# i 44 more rows
```

```
airports %>%
  group_by(STATE) %>%
  summarise(num_aeroportos = n()) %>%
  arrange(desc(num_aeroportos))
```

```
# A tibble: 54 × 2
  STATE num_aeroportos
  <chr>      <int>
1 TX          24
2 CA          22
3 AK          19
4 FL          17
5 MI          15
6 NY          14
7 CO          10
8 MN           8
9 MT           8
10 NC           8
# i 44 more rows
```

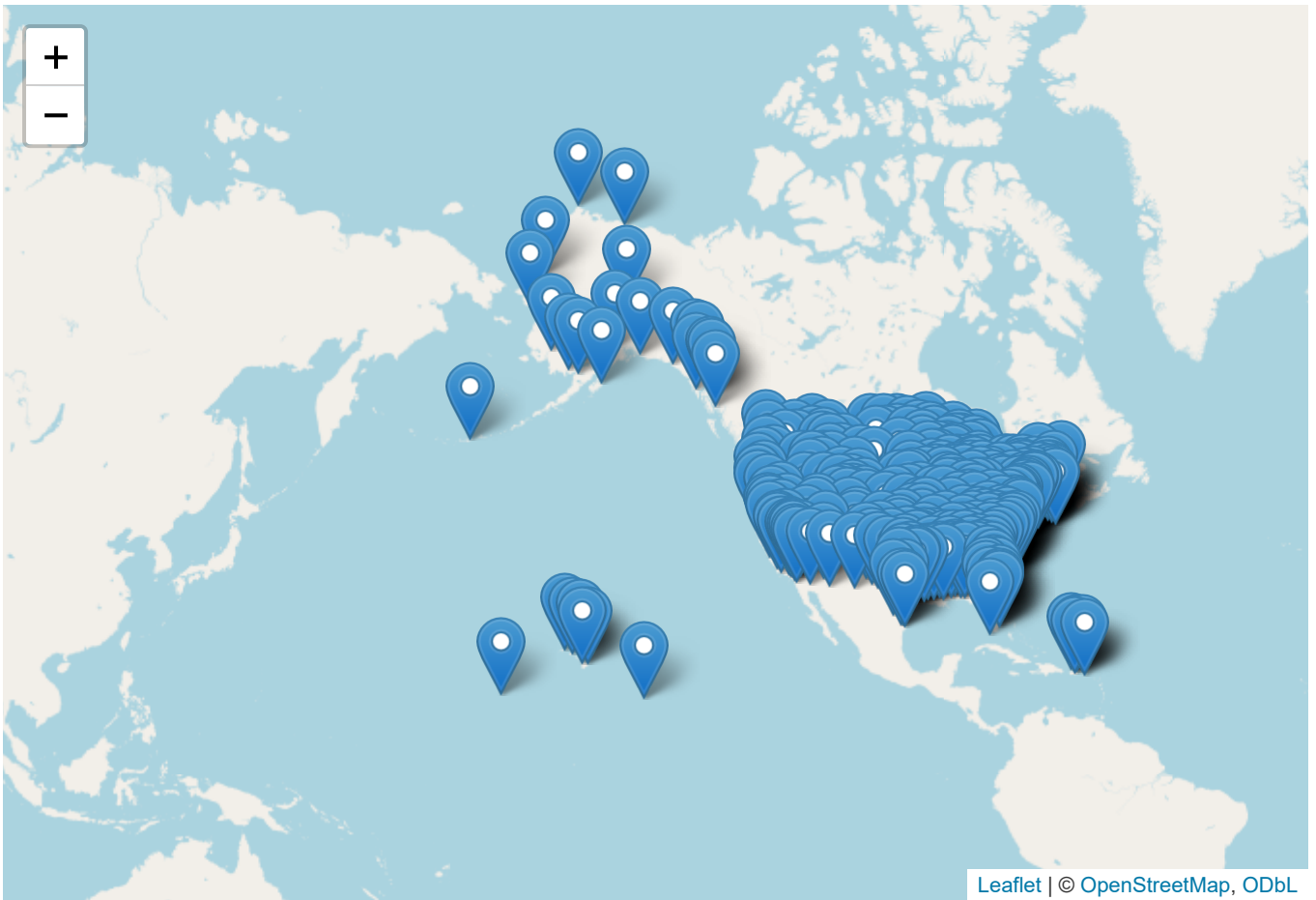
## Part 4

```
library(leaflet)
```

```
this <- flights %>%
  filter(!is.na(LATITUDE) & !is.na(LONGITUDE)) %>%
  leaflet() %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      "<b>Aeroporto:</b> ", DESTINATION_AIRPORT, "<br>",
      "<b>Cidade/UF:</b> ", CITY, " - ", STATE, "<br>",
      "<b>Atraso médio:</b> ", round(ARRIVAL_DELAY, 2), " min"
    )
  )
```

)

```
# mapa  
this
```



```
library(webshot2)
```